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AIML – B1

Theory of PCA (Principal Component Analysis)

Principal Component Analysis (PCA) is a statistical technique used for dimensionality reduction. It transforms a dataset containing many possibly correlated variables into a smaller set of new variables called **principal components**.
The main objective of PCA is to reduce the number of dimensions while preserving as much variance (information) from the original dataset as possible. It simplifies complex datasets, improves computational efficiency, and helps in visualizing high-dimensional data.

Key Concepts

1. Variance

PCA assumes that directions with higher variance carry more important structural information about the dataset. The first principal component captures the maximum possible variance, the second captures the next highest variance, and so on.

2. Orthogonal Components

Each principal component is orthogonal (perpendicular) to the others. This ensures:

- No redundancy between components
- No multicollinearity
- Independent representation of information

3. Eigenvalues and Eigenvectors

PCA is mathematically based on eigenvalues and eigenvectors of the covariance matrix of the data.

- **Eigenvectors** determine the direction of maximum variance (principal components).
- **Eigenvalues** indicate how much variance is captured by each principal component.

Components with larger eigenvalues retain more information and are usually selected for dimensionality reduction.

4. Covariance Matrix

PCA begins by computing the covariance matrix to understand relationships between features. If features are highly correlated, PCA can effectively combine them into fewer components.

Steps Involved in PCA

1. Standardize the dataset (important when features have different scales).
2. Compute the covariance matrix.
3. Calculate eigenvalues and eigenvectors.
4. Sort eigenvectors based on descending eigenvalues.
5. Select the top k principal components.
6. Transform the original data into the new reduced feature space.

Example of PCA

Consider a dataset of students with the following features:

- Study hours per week
- Number of practice tests taken
- Final exam score

These features are likely correlated. For example:

- Students who study more hours usually take more practice tests.
- Students who study more tend to score higher on exams.

Since these variables are related, there is redundancy in the dataset.

After applying PCA:

- Principal Component 1 may represent overall academic effort (combining study hours and practice tests).
- Principal Component 2 may represent performance variation not fully explained by effort.

Instead of using three correlated features, the model can use two principal components that capture most of the information.

This reduces dimensionality while preserving the overall structure of the data.

Advantages of PCA

- Reduces dimensionality
- Removes multicollinearity
- Reduces computational cost
- Improves visualization of high-dimensional data
- Helps prevent overfitting in high-dimensional datasets

Limitations of PCA

- PCA does not consider class labels (unsupervised method).
- Principal components are less interpretable than original features.

- Important discriminative information may sometimes be lost.
- Performance improvement is not guaranteed for low-dimensional datasets.

Result Analysis from datasets

1. Iris Dataset

In the Iris dataset, PCA showed that the first principal component alone captures approximately 72.77% of the total variance, while the first two components together explain about 95.8% of the variance. This indicates that most of the information in the dataset can be represented in just two dimensions instead of four. Since the dataset is already low dimensional and well structured, PCA mainly helps in visualization rather than improving model performance. The variance distribution confirms that the dataset has strong feature correlations but does not require aggressive dimensionality reduction.

2. Breast Cancer Dataset

For the Breast Cancer dataset, the model trained without PCA achieved an accuracy of **0.9825**, indicating strong predictive performance using all 30 features.
When dimensionality reduction was applied:

- Using only PC1 (43.17% variance retained) reduced accuracy to **0.9240**, showing that one component alone is insufficient.
- Using PC1 and PC2 (≈63% variance retained) improved accuracy to **0.9766**, which is very close to the original model.
- Using PC1, PC2, and PC3 further retained more variance and produced accuracy of **0.9708**.

This demonstrates that PCA effectively reduces dimensionality while preserving most classification power when sufficient components are retained. However, excessive reduction (like using only PC1) leads to loss of discriminative information.

Overall Conclusion

The experimental results show that PCA compresses correlated features into fewer orthogonal components while retaining maximum variance. In the Iris dataset, two components were sufficient to represent nearly the entire dataset variance, highlighting its low-dimensional nature.
In contrast, the Breast Cancer dataset benefited more from PCA due to its higher dimensionality and feature correlations. While the original accuracy was highest without reduction (0.9825), using two principal components achieved nearly similar performance (0.9766), proving that dimensionality can be reduced significantly without major loss in predictive ability.

Thus, PCA is more impactful and meaningful for higher-dimensional datasets, especially when features are highly correlated, whereas in low-dimensional datasets its benefit is primarily structural rather than performance-based.